# Gateways

# Gateways

- Gateways are BPMN flow elements, which are used to control how sequence flows interact as they converge and diverge within a process.
- Gateways enable the implementation of branching, forking, joining and merging of the paths in a business process diagram.
- In order to define different process flow behaviors, BPMN provides different types of Gate  Camunda Gateways and XOR gateway ways.
- These are defined with different icons within the diamond shape of the Gateway.
- Totally, BPMN 2.0 defines seven different types of Gateways.

# Gateways

**Types of Gateways**

- **Data-based Exclusive Gateway (XOR)**
  - Exclusive gateway is a diversion point of a business process flow. For a given instance of the process, there is only one of the paths can be taken
- **Parallel Gateway**
  - Model fork / join concurrency.
- **Inclusive Gateway**
  - Model conditional fork / join concurrency.
- **Event-based Gateway**
  - Model decisions based on events.

# Gateways

**Data-based Exclusive Gateway (XOR)**

**Exclusive gateway:**

- An exclusive gateway (also called the XOR gateway or, in more technical terms, the exclusive data-based gateway), is used to model a decision in the process.
- When the execution arrives at this gateway, all outgoing sequence flows are evaluated in the order in which they have been defined.
- The sequence flow which condition evaluates to 'true' (or which doesn't have a condition set, conceptually having a 'true' value defined on the sequence flow) is selected for continuing the process.

# Gateways

**Data-based Exclusive Gateway (XOR)**

**Only one sequence flow:**

- Note that only one sequence flow is selected when using the exclusive gateway.
- In case multiple sequence flow have a condition that evaluates to 'true'.

**Runtime exception:**

- If no sequence flow can be selected (no condition evaluates to 'true') this will result in a runtime exception, unless you have a default flow defined.
- One default flow can be set on the gateway itself in case no other condition matches - like an 'else' in programming languages.

# Gateways

**Parallel gateway**

- A Parallel Gateway or AND Gateway is used to synchronize (combine) parallel flows and to create parallel flows
- A Parallel Gateway creates parallel paths without checking any conditions; each outgoing Sequence Flow receives a token upon execution of this Gateway
- For incoming flows, the Parallel Gateway will wait for all in Camunda Parallel Gateway coming flows before triggering the flow through its outgoing Sequence Flows

# Gateways

**Inclusive OR gateway**

- The Inclusive Gateway can be seen as a combination of an exclusive and a parallel gateway.
- Like an exclusive gateway, you can define conditions on outgoing sequence flows and the inclusive gateway will evaluate them.
- However, the main difference is that the inclusive gateway can receive more than one sequence flow, like a parallel gateway.
- A diverging Inclusive Gateway (Inclusive Decision) can be used to create alternative but also parallel paths within a Process flow

# Gateways

**Inclusive OR gateway**

- Unlike the Exclusive Gateway, all condition Expressions are evaluated
- The true evaluation of one condition Expression does not exclude the evaluation of other condition Expressions
- All Sequence Flows with a true evaluation will be traversed by a token
- Since each path is considered to be independent, all combinations of the paths MAY be taken, from zero to all. However, it should be designed so that at least one path is taken.

# Gateways

**Inclusive OR gateway**

- The functionality of the inclusive gateway is based on the incoming and outgoing sequence flows:
  - **fork:** all outgoing sequence flow conditions are evaluated and for the sequence flow conditions that evaluate to 'true', the flows are followed in parallel, creating one concurrent execution for each sequence flow.
  - **join:** all concurrent executions arriving at the inclusive gateway wait at the gateway until an execution has arrived for each of the incoming sequence flows that have a process token. This is an important difference to the parallel gateway. So in other words, the inclusive gateway will only wait for the incoming sequence flows that are executed. After the join, the process continues past the joining inclusive gateway.